

ClassRoomSTORM Theory Notes

Diffraction, Blinking, and the Physics of Localization Microscopy

Awanish Pratap Singh

Companion to ClassRoomSTORM Studio, software version 1.1

Abstract

These notes explain the physics behind ClassRoomSTORM Studio V1.1. They build, step by step, from the diffraction limit of an ordinary microscope to the idea of localization microscopy: why a blurred spot can still be *located* far more precisely than it can be *resolved*, and how many such locations accumulate into a sharp image. The central message is a distinction students should carry away: *resolution* — how finely an instrument separates two points in a single snapshot — is set by optical blur, whereas *localization precision* — how well the centre of one isolated spot is known — is a statistical quantity set by photon counts, background, and noise. The final sections describe exactly how ClassRoomSTORM V1.1 reconstructs an image, and state honestly what the centroid method does and does not do. The V1.1 release adds the unified Studio launcher while retaining that transparent teaching reconstruction method.

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1 Introduction: The Resolution Problem

A microscope cannot show arbitrarily fine detail. Two points of light that are close enough together blur into a single smudge, no matter how good the lens. This limit — the *diffraction limit* — has been understood since the nineteenth century [1] and sits at roughly half the wavelength of light, about 200–250 nm for visible light [21, 13].

For more than a century this was considered an unbreakable barrier. Modern *super-resolution* microscopy sidesteps it. One especially elegant family of methods — *single-molecule localization microscopy* (SMLM), which includes STORM and PALM — does so with a simple statistical trick: never let two nearby emitters be visible at the same time. If only a few, well-separated emitters glow in any one frame, each appears as an isolated blurred spot, and the *centre* of an isolated spot can be estimated very precisely. Repeating this over thousands of frames and accumulating the centres yields a far sharper picture than any single frame [16, 15].

ClassRoomSTORM is a teaching application that reproduces the *statistical and computational logic* of SMLM. Its *Virtual Experiment* generates blinking videos with controllable blur, brightness, and noise; its *SR Reconstruction* workflow locates the spots and accumulates them. These notes explain the physics this software embodies. They are meant to be read alongside the *ClassRoomSTORM Help* document, which explains the buttons and screens.

ClassRoomSTORM is a learning tool. It is not a fluorescence microscope, and its reconstruction uses a deliberately simple, transparent estimator rather than a research-grade one. Throughout these notes, material describing the *ideal* statistical method is clearly marked in violet boxes; the running text and section 10 describe what ClassRoomSTORM V1.1 actually does.

2 The Diffraction Limit

When light from a single point passes through a lens of finite size, diffraction spreads it into a small pattern of rings called the *Airy pattern*:

$$I(r) = I_0 \left[\frac{2J_1(\alpha r)}{\alpha r} \right]^2, \quad \alpha = \frac{2\pi \text{NA}}{\lambda}, \quad (1)$$

where r is the radial distance from the centre, I_0 is the peak intensity, J_1 is the first-order Bessel function, λ is the wavelength of light, and NA is the *numerical aperture* of the lens (a measure of how wide a cone of light it collects). A perfect point of light is therefore never imaged to a point — it is imaged to a small bright disc.

Two points can only be told apart if their discs do not overlap too much. Abbe’s criterion gives the smallest resolvable separation [1, 21]:

$$d_{\text{Abbe}} \simeq \frac{\lambda}{2\text{NA}}. \quad (2)$$

A closely related statement, the Rayleigh criterion, places the threshold where the peak of one Airy disc falls on the first dark ring of the other [21, 18]:

$$d_{\text{Rayleigh}} \simeq \frac{0.61 \lambda}{\text{NA}}. \quad (3)$$

For green light, $\lambda \approx 550$ nm, and a good objective, $\text{NA} \approx 1.4$. Then $d_{\text{Abbe}} \approx 550/(2 \times 1.4) \approx 196$ nm and $d_{\text{Rayleigh}} \approx 0.61 \times 550/1.4 \approx 240$ nm. This is the few-hundred-nanometre barrier that super-resolution methods must overcome.

Figure 1 shows how two emitters merge as their separation shrinks toward this limit. The key point for what follows: eqs. (2) and (3) describe *resolution* — the ability to separate two points that are visible *at the same time*. They say nothing about how well the position of a *single, isolated* point can be measured. That is a different question, and it has a much more optimistic answer.

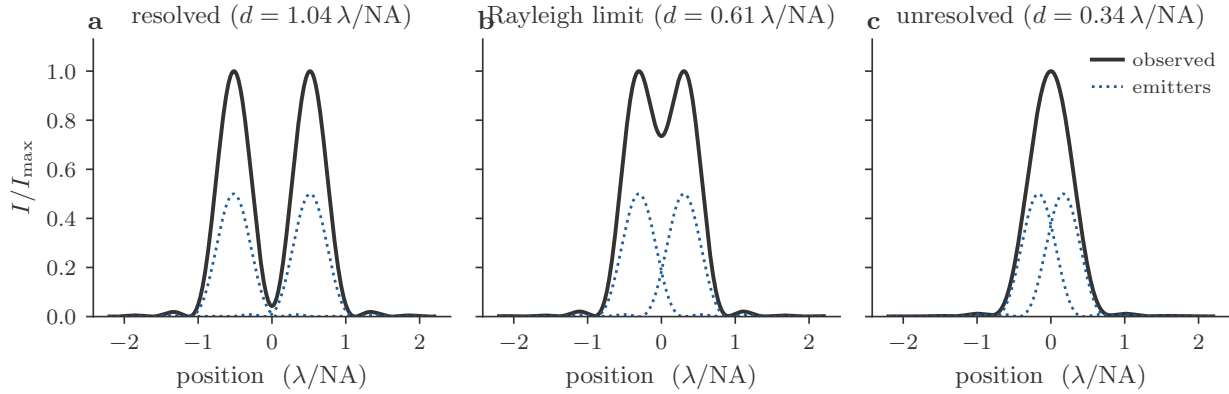


Figure 1: The diffraction limit. Two emitters imaged as Airy point-spread functions: clearly resolved when well separated (left), marginally resolved at the Rayleigh separation (centre), and merged into a single blur when too close (right).

3 The Point Spread Function

The image of a single point source is called the *point spread function* (PSF). The Airy pattern of eq. (1) is the exact PSF of an ideal lens, but for calculations a simpler model is almost always used: a two-dimensional Gaussian [6, 10]. An isolated emitter is modelled as

$$I(x, y) = b + \frac{N}{2\pi\sigma^2} \exp\left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2}\right], \quad (4)$$

where (x_0, y_0) is the true emitter position, N is the total number of signal photons, b is a background level, and σ is the *width* of the PSF. The width is set by the optics and the pixel size a of the camera; a useful approximate relation is [6]

$$\sigma \approx 0.21 \frac{\lambda}{\text{NA} \cdot a}. \quad (5)$$

The Gaussian is not exact — it ignores the Airy rings — but it captures the central bump well (fig. 2) and makes the algebra of localization tractable.

In ClassroomSTORM, the *Virtual Experiment* renders each blinker as a Gaussian spot, and you set its width directly with the `sigma_px` control. Choosing a large σ is the same as building a more strongly diffraction-limited “microscope.”

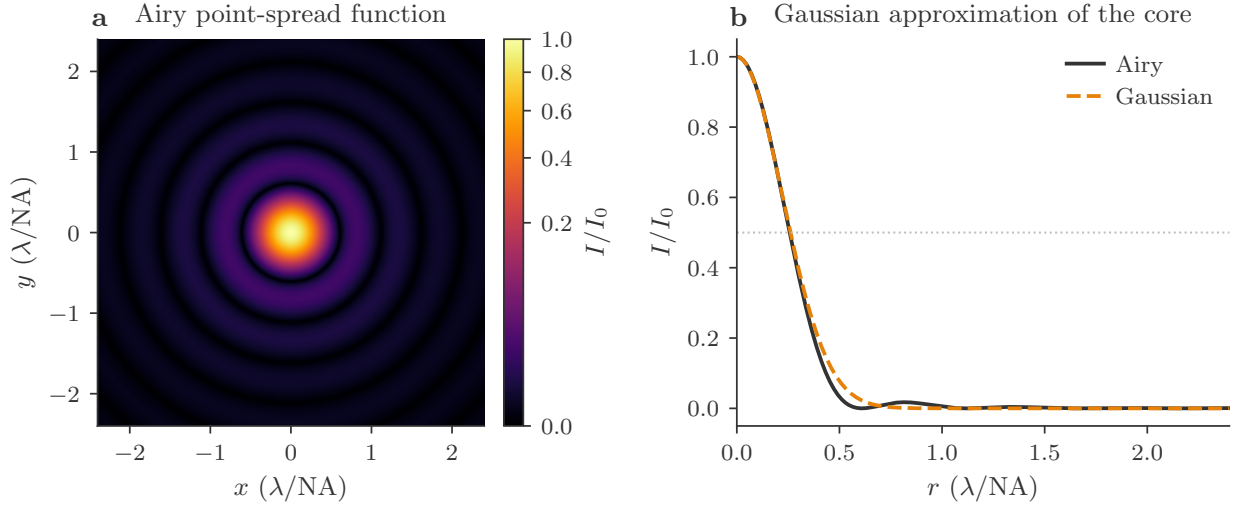


Figure 2: (a) The Airy point-spread function of an ideal lens. (b) Its central lobe is well approximated by a Gaussian; the rings are neglected.

4 Image Formation and Sampling

A real image is the true scene blurred by the PSF and then corrupted by noise. Writing the true distribution of light as $f(x, y)$, the PSF as $h(x, y)$, and the noise as $\eta(x, y)$, the recorded image is

$$g(x, y) = (h * f)(x, y) + \eta(x, y), \quad (6)$$

where $*$ denotes convolution. Convolution by h is exactly the blurring of section 2: every point in f is replaced by a copy of the PSF, as fig. 3 illustrates.

A camera does not sample g at infinitely fine points. Each pixel *collects all the light that lands on its small square*. The expected number of counts in the pixel at column j , row i , for a single Gaussian emitter, is the Gaussian *integrated* over that pixel [7]:

$$M_{ij} = b + N G_x(j; x_0, \sigma) G_y(i; y_0, \sigma), \quad (7)$$

$$G_x(j; x_0, \sigma) = \frac{1}{2} \left[\operatorname{erf} \frac{j + \frac{1}{2} - x_0}{\sqrt{2} \sigma} - \operatorname{erf} \frac{j - \frac{1}{2} - x_0}{\sqrt{2} \sigma} \right], \quad (8)$$

with G_y defined the same way along the rows. The factor G_x is simply the fraction of the Gaussian's light that falls within the width of pixel j ; erf is the error function. Equations (7) and (8) are the honest *forward model*: they say what the camera *should* record, on average, for an emitter at (x_0, y_0) . Localization, in section 7, is the problem of running this model backwards.

5 Photon Noise

Light arrives as discrete photons at random times. If a pixel expects M_{ij} counts on average, the number it *actually* records, Z_{ij} , fluctuates according to Poisson statistics [8]:

$$Z_{ij} \sim \text{Poisson}(M_{ij}). \quad (9)$$

A defining property of the Poisson distribution is that its variance equals its mean. Real cameras add a second, electronic noise (*read noise*) on top, so the total per-pixel variance is approximately [14]

$$\text{Var}[Z_{ij}] \approx M_{ij} + \left(\frac{\sigma_{\text{read}}}{g} \right)^2, \quad (10)$$

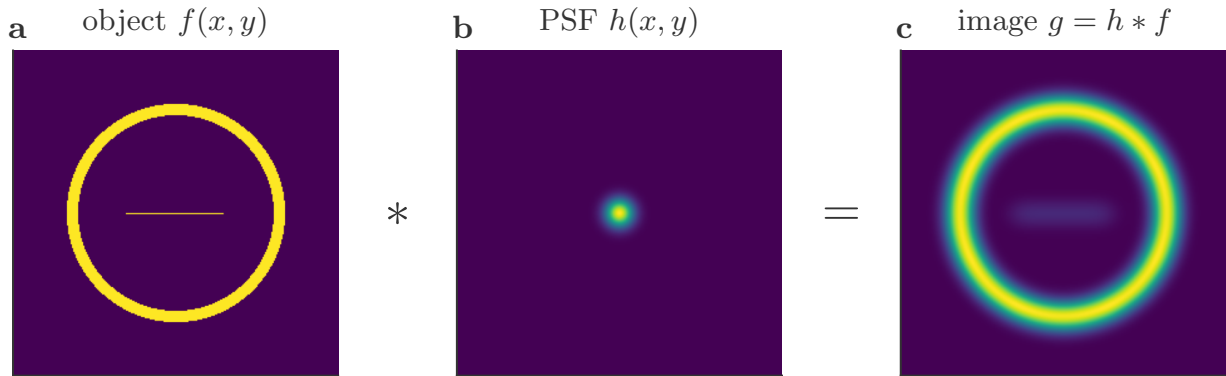


Figure 3: Image formation as a convolution: the true object f is blurred by the point-spread function h to give the recorded image $g = h * f$.

where σ_{read} is the camera read noise and g its gain.

Photon noise is not a nuisance to be ignored; it is the fundamental reason localization has a *precision limit* (section 7.2). The more photons an emitter delivers, the smaller the relative fluctuations, and the more precisely its position can be pinned down. Figure 4 shows how repeated measurements of the same emitter scatter because of this noise. In ClassroomSTORM, the **Brightness** control sets the photon budget N and **Read noise** sets σ_{read} — so students can watch precision improve as they turn the brightness up.

6 Blinking and Single-Molecule Localization Logic

Now the central idea. Suppose a sample contains many emitters packed more closely than the diffraction limit. If they all glow at once, their PSFs overlap and eq. (6) produces an unresolvable blur — the diffraction limit wins (fig. 5a).

But suppose instead that the emitters *blink*: at any instant only a small, random subset is switched on, and those that are on happen to be far apart. Then each frame contains only a few *isolated* spots (fig. 5b). Each isolated spot can be located precisely (section 7). Over thousands of frames, different emitters take their turn, and the accumulated set of locations traces out the whole structure.

This is single-molecule localization microscopy. STORM [2] and PALM [3], both introduced in 2006, use photoswitchable dyes and fluorescent proteins to make molecules blink; the same logic underlies the broader field of far-field optical nanoscopy [5, 14]. It is worth noting that the diffraction limit is never actually *broken*: it is *sidestepped*, by arranging that two unresolved emitters are never on at the same time. (Other super-resolution methods, such as STED [4], take a different route by physically sharpening the PSF.)

ClassroomSTORM reproduces this logic. The *Virtual Experiment* builds exactly such blinking videos: you choose how many emitters may be on per frame and how far apart they must be. The companion macroscopic experiment uses blinking LEDs in place of fluorophores — emitters that are large in physical size but produce data with the *same statistical structure* as SMLM. Low-cost real SMLM instruments built in the same spirit have been described in the literature [19, 20].

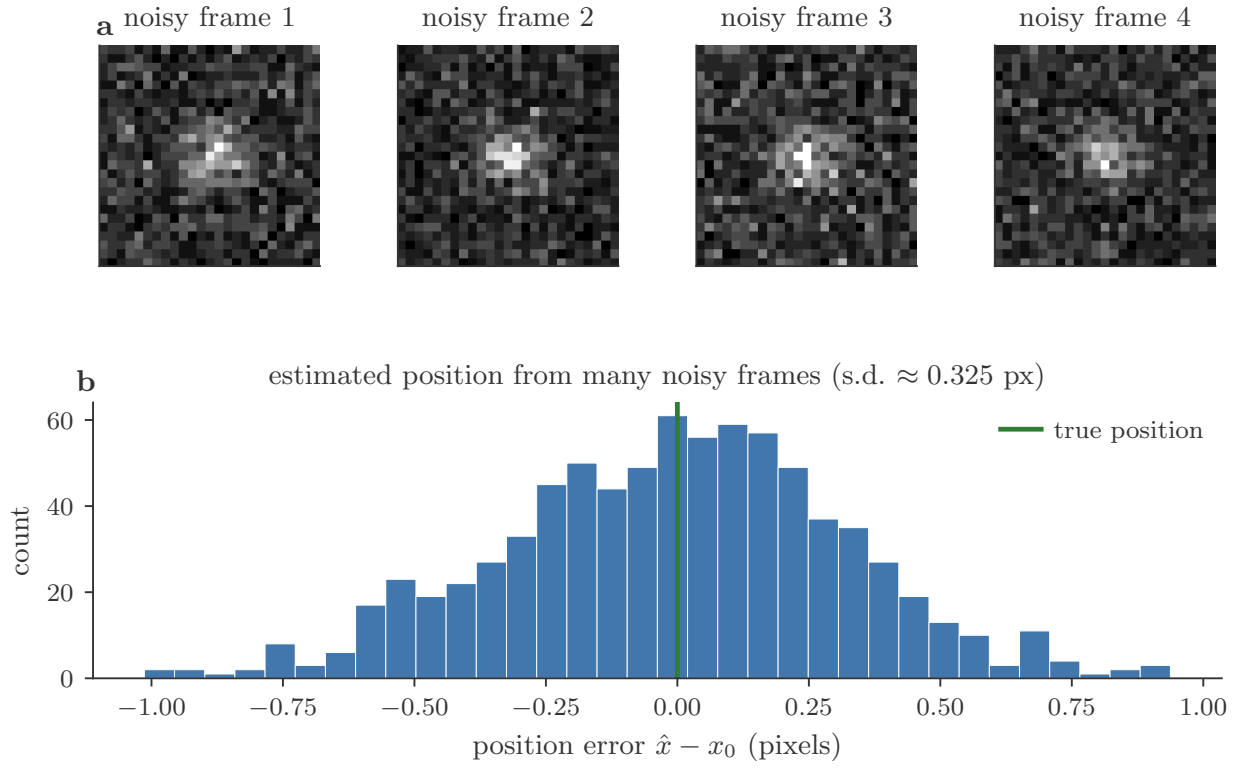


Figure 4: Photon noise. The same emitter recorded several times gives different noisy frames (top); the position estimated from many such frames scatters about the true value (bottom).

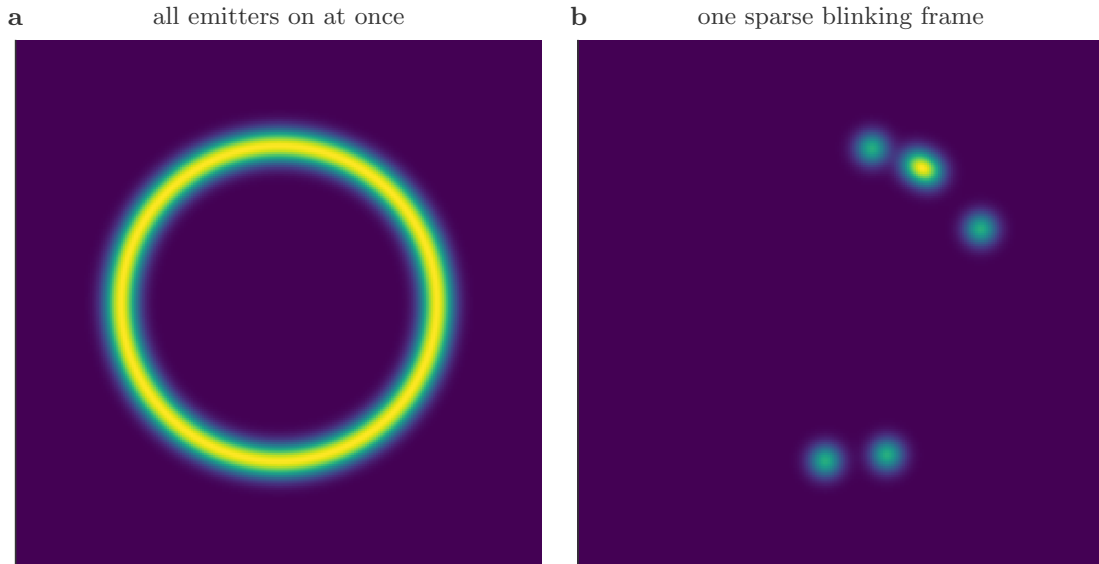


Figure 5: Why blinking helps. (a) With all emitters on at once their point-spread functions overlap into an unresolvable blur. (b) In a single sparse frame only a few emitters are on, and each appears as an isolated spot that can be located precisely.

Localization works *only* when spots are isolated. If two emitters are on together and their PSFs overlap, the methods in section 7 return one position somewhere between them — a wrong answer, not two right ones. This is why the *Virtual Experiment* lets you control `Max blinkers/frame` and `Blinker spacing`, and why crowded data reconstructs poorly.

7 Localizing an Emitter

Given one isolated, blurred spot, where is its centre? This section gives the simple, transparent answer ClassroomSTORM V1.1 uses, the precision that answer can reach, and — as clearly marked background — the statistically optimal answer.

7.1 The centroid estimator used in V1.1

ClassroomSTORM V1.1 localizes with the *intensity-weighted centroid* — the “balance point” of the bright pixels [10]. First the frame is *thresholded*: only pixels brighter than a high percentile of the frame (the top 0.5%) are kept, forming a mask $\mathcal{M}$ of candidate spot pixels. The centre estimate is then the brightness-weighted average of the masked pixel coordinates:

$$\hat{x}_0 = \frac{\sum_{(i,j) \in \mathcal{M}} x_j Z_{ij}}{\sum_{(i,j) \in \mathcal{M}} Z_{ij}}, \quad \hat{y}_0 = \frac{\sum_{(i,j) \in \mathcal{M}} y_i Z_{ij}}{\sum_{(i,j) \in \mathcal{M}} Z_{ij}}. \quad (11)$$

The centroid has two virtues for teaching: it needs no iterative fitting, and every step — threshold, mask, weighted average — is inspectable (fig. 6). Crucially, the estimate $(\hat{x}_0, \hat{y}_0)$ is a *continuous* number: it is not restricted to whole-pixel positions, so it can locate a spot to a fraction of a pixel even though the spot itself is several pixels wide.

Equation (11) uses the raw pixel values Z_{ij} and does *not* subtract a background estimate first. If the background b is appreciable, the centroid is pulled slightly toward the brighter side of the spot — a small bias. ClassroomSTORM V1.1 limits this by keeping only the brightest pixels through the high threshold. Subtracting a local background before eq. (11), and fitting the full model of eq. (7), are improvements discussed in section 12.

7.2 Localization precision (background)

Because of photon noise (section 5), repeating the measurement gives a slightly different $\hat{x}_0$ each time. The spread of those estimates is the *localization precision*, usually written as a standard error SE. A widely used closed-form estimate is the Thompson–Larson–Webb formula [6]:

$$\text{SE} \approx \sqrt{\frac{\sigma^2 + a^2/12}{N} + \frac{8\pi\sigma^4 b^2}{a^2 N^2}}, \quad (12)$$

where σ is the PSF width, a the pixel size, N the number of signal photons, and b the background per pixel. The formula has two terms:

- the **photon term** $(\sigma^2 + a^2/12)/N$ dominates when the signal is strong; it makes the precision improve as $1/\sqrt{N}$;
- the **background term** $8\pi\sigma^4 b^2/(a^2 N^2)$ dominates when the signal is weak relative to the background.

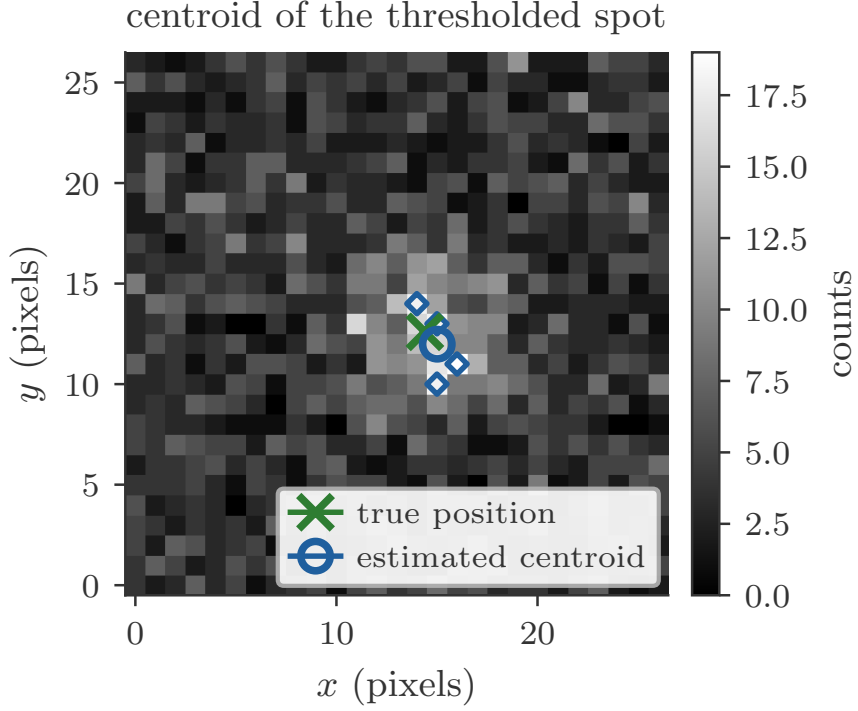


Figure 6: The V1.1 estimator: the frame is thresholded to a mask of the brightest pixels (blue outline), and the intensity-weighted centroid (eq. (11)) of those pixels is taken as the emitter position.

Figure 7 plots this behaviour.

Take $\sigma = 1.5$ px and a bright spot, $N = 10^4$ photons, with negligible background. The photon term gives $\text{SE} \approx \sqrt{(1.5^2)/10^4} \approx 0.015$ px — about one part in a hundred of the blur width. With only $N = 100$ photons, $\text{SE} \approx 0.15$ px. The emitter is blurred over several pixels, yet its *centre* is known to a tiny fraction of a pixel. This is the heart of localization microscopy: *resolution* (the blur width, eq. (2)) and *precision* (how well one centre is known, eq. (12)) are different quantities, and the second can be far smaller than the first.

7.3 The statistical ideal (background only): MLE and the Cramér–Rao bound

The centroid of eq. (11) is simple but not statistically optimal. The *best* estimate uses the full forward model of eq. (7). Because the data are Poisson-distributed (eq. (9)), the optimal estimator *maximizes the likelihood* of the observed counts, equivalently minimizing the negative log-likelihood

$$L(\theta) = \sum_{i,j} [M_{ij}(\theta) - Z_{ij} \ln M_{ij}(\theta)], \quad (13)$$

over the parameters $\theta = (x_0, y_0, N, b, \sigma)$. No estimator, however clever, can do better than

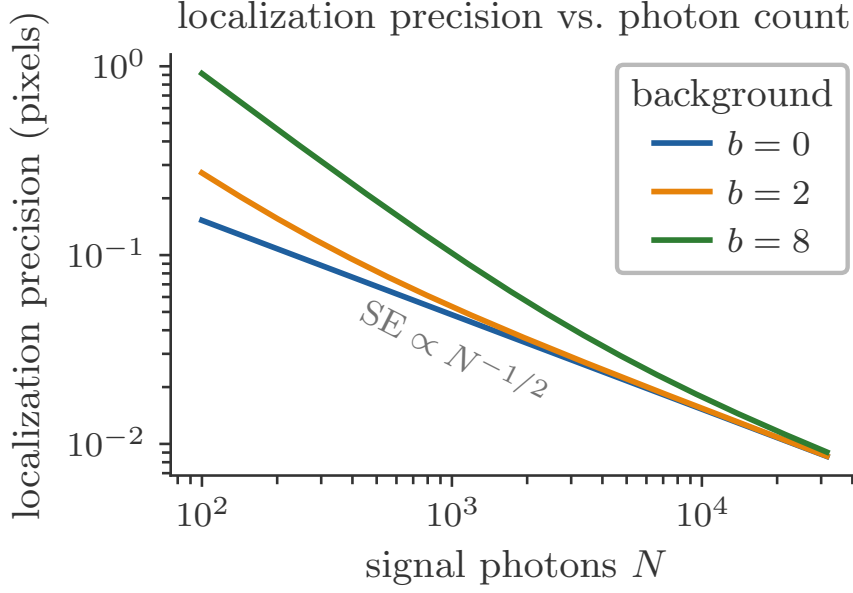


Figure 7: Localization precision predicted by the Thompson formula (eq. (12)). Collecting more photons sharpens the position estimate; in the photon-limited regime the precision improves as $1/\sqrt{N}$.

the *Cramér–Rao lower bound* [8, 7]:

$$\text{Var}(\hat{\theta}) \geq F^{-1}(\theta), \quad F_{mn} = \sum_{i,j} \frac{1}{M_{ij}} \frac{\partial M_{ij}}{\partial \theta_m} \frac{\partial M_{ij}}{\partial \theta_n}, \quad (14)$$

where F is the Fisher information matrix. Maximum-likelihood fitting attains this bound; a properly weighted least-squares fit comes very close. ClassroomSTORM V1.1 uses the simpler centroid *on purpose* — to keep every step visible — and so sits a little above this ideal floor. It does *not* implement eq. (13) or eq. (14); these are background theory and a possible extension (section 12). Equation (12) is, in effect, a convenient approximation to eq. (14) for the centre coordinate.

8 Precision versus Accuracy

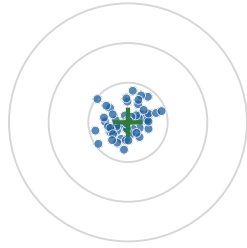
Two estimates can be tightly clustered yet systematically wrong. It is essential to keep two ideas separate [9]:

Precision is *repeatability* — how closely repeated measurements of the same emitter agree with one another. It is the random scatter, and it is what eq. (12) estimates.

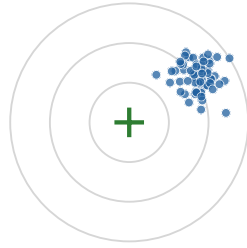
Accuracy is *correctness* — how close the estimates are to the *true* position. A constant offset (a bias) hurts accuracy without hurting precision.

Figure 8 contrasts the two. The background bias of the V1.1 centroid (section 7.1) is exactly an accuracy problem: it can shift estimates without widening their scatter. This is why ClassroomSTORM’s *Virtual Experiment* saves the true positions in `truth.csv`: comparing reconstructed positions against the truth measures *accuracy*, which precision alone cannot reveal.

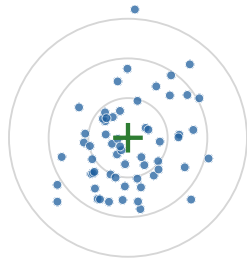
a high precision, high accuracy



b high precision, low accuracy



c low precision, high accuracy



d low precision, low accuracy

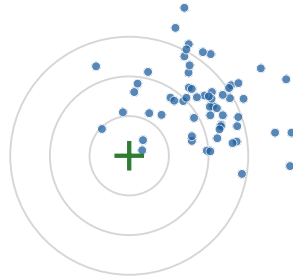


Figure 8: Precision versus accuracy, illustrated for four cases (panels **a–d**). Each blue cluster is repeated localizations of one emitter; the green cross is its true position. Precision is the tightness of the scatter, accuracy is its closeness to the truth, and the two properties are independent.

9 From Localizations to a Reconstructed Image

A list of localized positions is not yet a picture. ClassroomSTORM *renders* it: each accepted localization is drawn as a small Gaussian “splat,” and all the splats are summed into the super-resolution image. In V1.1 every splat is drawn with the same fixed width (about 1.5 px); the splat width is a display choice that makes the sparse points visible, not a measurement.

A precise list of points is still not enough — there must be *enough* points. This is a sampling requirement, analogous to the Nyquist criterion: to portray a structure of a given fineness, localizations must be spaced no farther apart than about half that fineness [18, 14]. Equivalently,

$$(\text{structural resolution}) \gtrsim 2 \times (\text{localization precision}). \quad (15)$$

A sharp pencil is not enough to draw a fine figure; you also need enough dots, as fig. 9 shows. More blinking frames usually means more localizations and better sampling — provided the detections stay reliable.

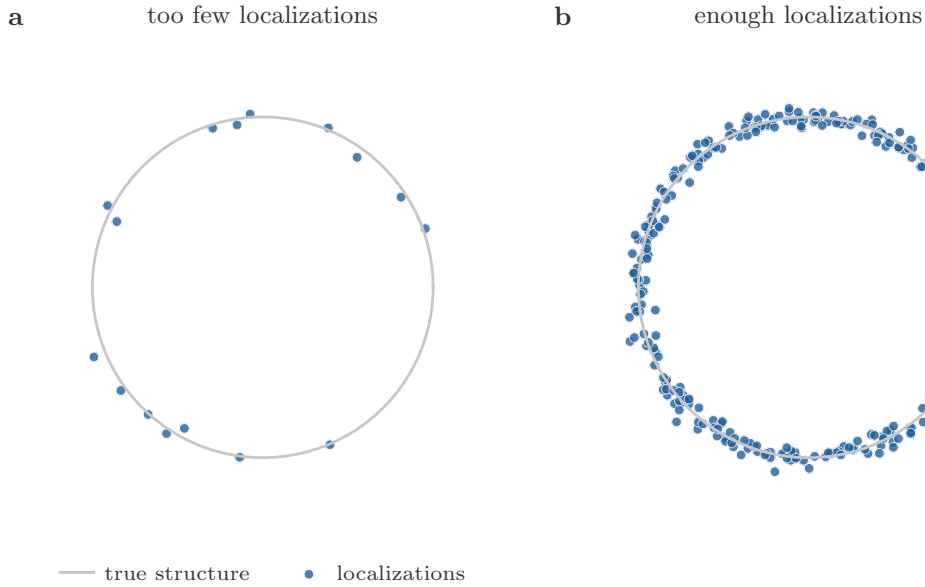


Figure 9: Sampling matters. Too few localizations leave the structure ambiguous (left); enough localizations reveal it (right), even though each individual localization is equally precise.

The width of a rendered splat is *not* the resolution of the result. Honest practice [18] is to report the localization precision (eq. (12)) and the sampling density (eq. (15)) separately, rather than quoting a single number. ClassroomSTORM is designed to make both visible rather than to print one headline figure.

10 The ClassroomSTORM V1.1 Pipeline

Putting the pieces together, the list below states exactly what ClassroomSTORM V1.1 does when you click *Run Reconstruction* (fig. 10). Each step is transparent and maps onto an equation above.

1. **Load** the blinking video frame by frame (optionally cropped).
2. **Threshold** each frame: pixels brighter than a high intensity percentile (the top 0.5%) form the candidate mask $\mathcal{M}$.

3. **Localize** with the intensity-weighted centroid of eq. (11) — once per frame in **single** mode, or once per separated bright region in **multiple** mode.
4. **Accumulate** the localizations from all frames.
5. **Render** each localization as a Gaussian splat ($\sigma \approx 1.5$ px) and sum them into the super-resolution image.
6. **Average** the frames separately to form the diffraction-limited *raw mean* image.
7. **Compare** the raw mean and the reconstruction, including with the linked profile viewer.

The *linked profile viewer* lets you compare the two images along any chosen line. Writing $I(x, y)$ for an image and (x_0, y_0) for the chosen point, the viewer plots the row and column profiles

$$I_x(x) = I(x, y_0), \quad I_y(y) = I(x_0, y), \quad (16)$$

for both the raw mean and the reconstruction. A narrow reconstruction peak beside a wide raw bump, on the same axes, is the visual proof that localization has beaten the blur — as fig. 11 demonstrates for a pair of emitter rows that the raw image cannot separate.

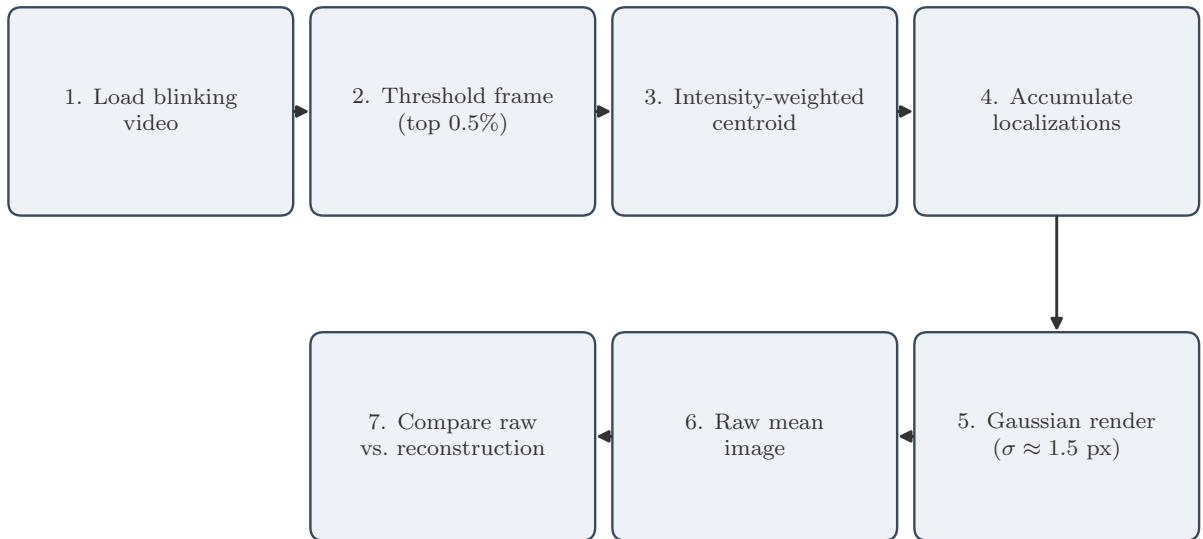


Figure 10: The ClassroomSTORM V1.1 reconstruction pipeline.

11 Scope, Assumptions, and Limitations

ClassroomSTORM V1.1 is honest about being a teaching tool. The limitations below are not defects to hide — each is a discussion point for students.

- **Overlapping emitters cannot be separated.** If two spots merge, the centroid of the merged region returns a single, wrong position (section 6). Reliable reconstruction needs sparse, separated blinking.
- **The centroid gives a position but no uncertainty.** V1.1 does not compute a

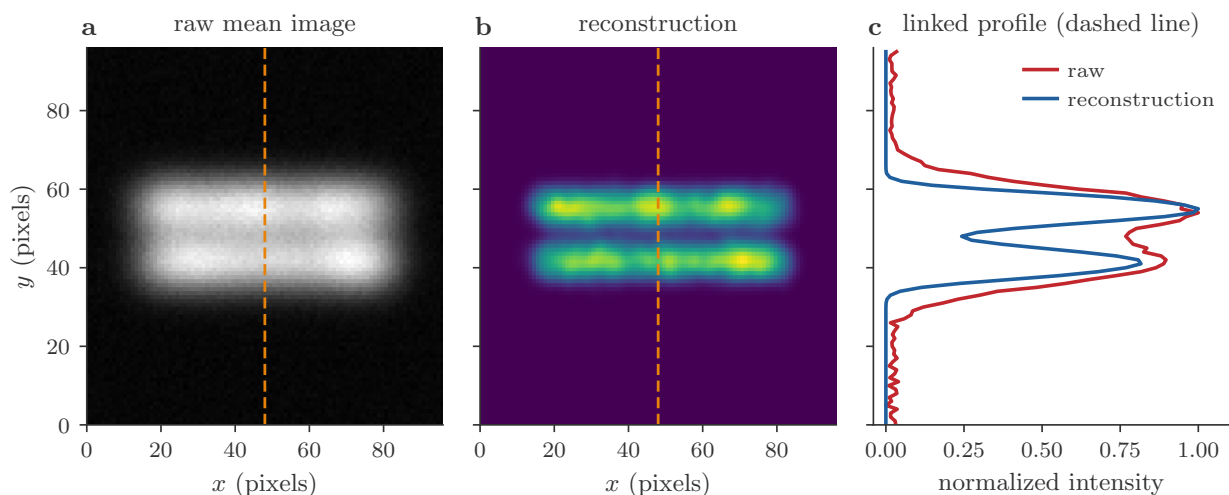


Figure 11: A reconstruction produced by the V1.1 pipeline on a simulated blinking video of two closely spaced emitter rows. The raw mean image blurs the rows together; the reconstruction separates them, and the linked profile shows one broad raw peak (red) versus two distinct reconstruction peaks (blue).

per-localization standard error. Equation (12) tells you what precision to *expect*; the software does not report it spot by spot.

- **No background subtraction before the centroid.** As noted in section 7.1, this can bias positions when the background is significant.
- **No model fitting.** V1.1 does not perform weighted-least-squares or maximum-likelihood Gaussian fitting; it does not evaluate the Cramér–Rao bound.
- **The render width is fixed.** Every splat is drawn at the same width; it shows where localizations are, not how precise each one is.
- **No drift correction and no applied calibration.** Results are reported in pixels; physical calibration is recorded but not applied.
- **Noisy, saturated, or under-exposed video reconstructs poorly,** because thresholding (section 5) can no longer separate signal from noise.

These are deliberate simplifications that keep every step transparent. They are the natural starting points for the extensions below.

12 Extensions and Further Reading

The V1.1 method can be improved in well-understood ways; these are standard in research SMLM and are natural project extensions. They are *not* part of ClassroomSTORM Studio V1.1.

- **Background subtraction** before the centroid (for example an annular median estimate) reduces the accuracy bias of section 7.1.
- **Gaussian fitting.** Replacing the centroid with a weighted least-squares or maximum-likelihood fit of the model in eq. (7) approaches the Cramér–Rao bound (eq. (14)) and yields a per-localization precision [7, 10].
- **Drift correction.** Slow stage drift can be measured by cross-correlating frames and subtracted before rendering [12].

- **Resolution metrics.** Fourier ring correlation gives an image-based resolution estimate from two independent halves of the data [18].
- **Reference software.** Research packages such as ThunderSTORM [11] implement the full pipeline and are useful for comparison. An accessible introduction to SMLM data analysis is given by Lee *et al.* [17], and the mathematical theory of recovering point sources is treated by Candès and Fernández-Granda [22].

A A Short Derivation: Why Precision Scales as $1/\sqrt{N}$

Derivation (optional)

A localization is, in essence, an average of the positions at which the N signal photons arrived. Each photon lands at a random position drawn from the PSF, which has a spread of order σ . The standard error of the mean of N independent samples, each with spread σ , is

$$\text{SE} \approx \frac{\sigma}{\sqrt{N}}.$$

This is the leading (photon-limited) term of the Thompson formula, eq. (12); the $a^2/12$ correction accounts for the finite pixel size, and the second term accounts for background photons that carry no position information. The lesson is simple and general: *collect more photons and the position sharpens, in proportion to the square root of the count.*

B Symbol Table

Symbol	Meaning
λ	Wavelength of light
NA	Numerical aperture of the imaging lens
$d_{\text{Abbe}}, d_{\text{Rayleigh}}$	Diffraction-limited resolution (eqs. (2) and (3))
h, f, g	Point spread function, true scene, recorded image (eq. (6))
σ	Width of the (Gaussian) point spread function
(x_0, y_0)	True position of an emitter; $(\hat{x}_0, \hat{y}_0)$ its estimate
N	Number of signal photons from an emitter
b	Background level per pixel
a	Camera pixel size
M_{ij}	Expected counts in pixel (i, j) (eq. (7))
Z_{ij}	Recorded counts in pixel (i, j)
σ_{read}, g	Camera read noise and gain
$\mathcal{M}$	Set of thresholded (candidate spot) pixels
SE	Localization precision (standard error)
$L(\theta), F$	Negative log-likelihood and Fisher information (eqs. (13) and (14))

Table 1: Symbols used in these notes.

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